

Fault-Based Test Case Prioritization of Regression Testing Using Genetic Algorithm

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ABSTRACT

Prioritization of test case technique schedules the test suite in an order which will increase the efficiency and effectiveness of the system. The test case is executed in an order which enhances the probability of finding the fault and also it finds the severe fault at the early stage. The prioritization of test case is proved to be useful for the regression testing. While much research is going on code coverage-based prioritization technique and requirement-based test case prioritization technique is not used by many. In this paper, test case prioritization technique based on fault of regression technique is proposed from the requirement specification to improve the software quality and to increase the fault detection rate. In this paper, test case prioritization is done based the six main factors: (1) customer priority, (2) changes made in requirement specification, (3) complexity in implementation of code, (4) re-usability, (5) flow of application, and (6) impact of the fault.

KEYWORDS

Average Number of Fault Detected, Genetic Algorithm, Regression Testing, Severe Faults, Test Case Prioritization

INTRODUCTION

In the software life cycle, the test case is written to test the software. If the software is altered, the maintenance activity called testing based on regression is performed to check the validity of the software that has been modified. After the modification of software, a new test case is added to the existing test suite for checking the altered requirements; thereby, it increases the size, cost required for the test suite (Jenifer and Illavarsi, 2014; Navak et al., 2017). To improve the process of testing, the software improvements have to be done in regression testing, which will reduce the cost required for testing. Once the requirements are specified, the software is developed with the help of a lifecycle in the development of software. Different phases are involved in the software development after designing the software; the testing section is used to test the developed software, a different method of testing techniques is used to check the efficiency of the code. For testing the whole software, regression testing is used. In the regression testing, the previously executed test case is also checked to check whether the previous code is working fine. The testing process is carried out to check whether fixing of new bugs does not result in new bugs which affect the previous bug-free functions (Sabharwal et al., 2011; Alsmadi and Alda, 2012). The regression testing is required (i) when there is a change in

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